

Use of Artificial Intelligence in Radiology: Impact on Pediatric Patients, a White Paper From the ACR Pediatric AI Workgroup

CME

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Abstract

In this white paper, the ACR Pediatric AI Workgroup of the Commission on Informatics educates the radiology community about the health equity issue of the lack of pediatric artificial intelligence (AI), improves the understanding of relevant pediatric AI issues, and offers solutions to address the inadequacies in pediatric AI development. In short, the design, training, validation, and safe implementation of AI in children require careful and specific approaches that can be distinct from those used for adults. On the eve of widespread use of AI in imaging practice, the group invites the radiology community to align and join Image IntelliGently (www.imageintelligently.org) to ensure that the use of AI is safe, reliable, and effective for children.

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INTRODUCTION

Artificial intelligence (AI) in medical imaging has rapidly expanded in recent years; however, very little AI has been dedicated to pediatric imaging. Furthermore, few products indicate whether they have been tested or validated in children. Of the >200 AI software as a medical device cleared by the FDA currently listed in the online ACR Data Science Institute catalog, only 6 (3%) indicate that they are intended for pediatric use [1]. These pediatric-use tools fall into the revised device classification for medical image management and processing systems. To date, there are no computer-aided detection, computer-assisted triage and notification software, or computer-aided diagnosis radiology products authorized for pediatric uses.

The practices of pediatric and adult radiology often differ widely, particularly with respect to age-related variations in physiology, anatomy, and diagnoses and differences in image acquisition parameters [2,3]. For example, pediatric images acquired with significantly reduced CT radiation doses, as recommended by the Image Gently® campaign [4], often have higher inherent image noise levels and may not be processed correctly by AI algorithms trained on higher dose adult images. An AI tool trained exclusively on adult patient data may not perform the same for the identification of pediatric disorders. Importantly, the FDA maintains different standards and controls for approval of products designed for children compared with adults. This process takes into account subpopulations on the basis of patient age and acknowledges that children's anatomy and physiology change with age [5].

The aims of this white paper are to educate the radiology community about the health equity issue of the lack of pediatric AI, improve the understanding of relevant pediatric AI issues, and offer solutions to address the inadequacies in pediatric AI development. In short, the design, training, validation, and safe implementation of AI in children require careful and specific approaches.

OVERVIEW OF CLINICAL DIFFERENCES FOR MEDICAL IMAGING OF PEDIATRIC PATIENTS

Children's health care needs differ from those of adults. For example, pediatric patient size varies widely, ranging from

premature neonates to adolescent bariatric patients. Normal changes in the size, shape, and appearance of multiple organs as well as the skeletal system over various ages as well as variations in patterns of injury and disease can present challenges for nonpediatric radiologists [6,7]. Examples include changes in the appearance of the thymus and bone marrow signal intensity over development as well as unique skeletal fracture patterns. Changes in pediatric physiology, such as variations in cerebral metabolism and organ perfusion and functional immaturity, also have implications for AI development and subsequent clinical performance [8,9]. Variations in vascular lumen size, cardiac output, and circulation time as well as end-organ perfusion state require different doses, concentrations, and administration techniques of intravascular contrast agents to produce the required tissue contrast enhancement [10]. An added challenge of imaging of pediatric patients is the frequent requirement of sedation or anesthesia, which also has implications for observed tissue perfusion [11]. Biometrics such as respiratory rate and cardiac output require consideration of age and are commonly used for respiratory and cardiac gating in image acquisition and accurate image interpretation.

It is widely recognized that children are more susceptible than adults to similar doses of ionizing radiation. Specifically, young children are more radiosensitive than adults, leading to higher cancer risk per unit dose of ionizing radiation [12,13]. The Image Gently campaign [4] successfully brought these issues to the forefront of physician consciousness, resulting in important changes to pediatric imaging protocols, particularly in North America. Therefore, exposure settings for diagnostic quality pediatric studies are now much lower than those used in adults at many institutions [14-16]. Pediatric radiologists may adjust pediatric protocol performance and interpretation, such as shortening or reducing the number of MR sequences, to decrease the duration of or altogether avoid the use of moderate sedation and anesthesia [17]. Therefore, assumptions about adult imaging protocols and tissue contrast may not apply to pediatric radiology.

Although there is overlap in some pediatric and adult pathologies, pediatric disorders have different prevalence and

characteristics than in adults, and some are exclusive to pediatric populations. Childhood cancers may also demonstrate different genetic profiles, histology, and behavior, resulting in different management strategies. All of these differences need to be taken into account in pediatric-specific AI development.

OVERVIEW OF CURRENT PEDIATRIC AI

The absence of commercially available AI algorithms for pediatric populations is multifactorial. Nearly all the publicly available data sets have exclusively included adult patients. Multiple data sets for the detection of pneumonia, cervical spine fractures, intracranial hemorrhage, brain tumors, pulmonary emboli, lung cancer, and COVID-19 pneumonia have been accumulated for patients >18 years of age [18]. In addition, a long-standing shortage of pediatric radiology subspecialists challenges access to expertise for requisite image labeling to support the development of AI algorithms. Such logistic challenges reduce feasibility and drive up development costs for vendors, impeding the development of software as a medical device for pediatric use. Furthermore, AI pricing and licensing are often volume based; therefore, the relative lower volume of pediatric studies may result in reduced revenue for pediatric AI developers.

Moreover, children are a protected population, with added regulatory requirements for conducting research on pediatric patients [19]. This creates specific barriers in bringing pediatric AI into the clinical environment. We are not aware of widely accepted legal and ethical standards addressing AI development and use in pediatric populations. As a result of the complex nature of pediatric consent, institutional legal teams are challenged to make decisions regarding intellectual property and data ownership issues for pediatric patients.

The generalization of adult algorithms to the pediatric population also presents challenges for pediatric patient safety, as institutions must weigh the benefits against the risks of what essentially represents off-label use. A few studies investigating the use of adult AI in pediatric patients demonstrated challenges with the generalizability of AI across the age spectrum. For example, AI developed to detect vertebral body fractures from adult patient data was neither reliable nor sensitive for vertebral body fracture detection in children [20,21].

In another investigation, the application in children of commercially available AI intended for adult chest radiography reported an overall accuracy of 87.5% in the analysis of 2,273 pediatric chest radiographs. Accuracy increased to 96.9% when patients younger than 2 years and those with cardiomegaly, corresponding to 33.3% of the original data set, were excluded. This investigation highlights the need for

detailed subgroup analysis when evaluating the performance of AI developed for adults in children [22]. Another recent study tested the ability of an adult algorithm to discriminate between benign and malignant thyroid nodules in children [23]. The specificity for pediatric malignant thyroid nodule classification was a mere 36.1%, emphasizing the lack of generalizability of AI classification performance with regard to pediatrics. In conclusion, the few available comparative pediatric and adult AI performance studies indicate a lack of straightforward generalizability to the pediatric patient population.

ETHICAL AND REGULATORY CONSIDERATIONS

The use of AI in pediatric radiology is associated with ethical and demographic concerns. Currently, most data used in AI development waive patient consent under institutional review board exemptions for retrospective studies. Data-sharing agreements assume the complete anonymization of images so that patients cannot be reidentified. In pediatric cases in which parental consent was obtained during standard of care treatment, additional issues arise. Should patients be able to opt out of continued inclusion in databases after reaching legal age [24]? Also, as a variety of statutes require retention of pediatric imaging until a child reaches adulthood, should imaging examinations that were previously interpreted without AI be reanalyzed when AI becomes available [25]? Moving forward, radiologists have an opportunity to address these questions and define how the combination of their native work plus AI should be applied to clinical practice. The complexity of guidelines for the need for signed consent before the use of AI in clinical care (ie, in a paradigm in whom management decisions are made essentially on AI performance) will need special attention, such as with caregiver acceptance during the consent and assent in children.

The ACR communicates openly and frequently with the FDA and other regulatory bodies regarding AI policies relevant to radiology patient care. The ACR's November 15, 2021, comments on AI transparency merit specific discussion as directly relevant to the pediatric population [26]. In that comment submission, the ACR advocated for information about the data sets used to develop and test AI devices to be publicly accessible and understandable, such as in a searchable public database. The ACR comment submission listed all information that health care providers should be able to access in such a database to guide decision making related to AI device acquisition, implementation, and appropriate use.

For pediatric patients, transparency on population demographics is of particular importance because AI that is

developed for adults may not be generalizable to children. The ACR also recommended including a “statement regarding authorization specifically for use in children, including a description of the evidence that does/does not support use in children, or if there is a lack of such evidence” on all FDA-authorized AI devices. Furthermore, the ACR recommended a centrally accessible and highly visible nutrition label-style summary of AI-enabled device information that could include a “pediatric use section” and would serve to “simplify pertinent data that may otherwise be tucked away in complex regulatory documents.” This would empower consumers, purchasers, and end-users within health care facilities to make better informed purchasing and medical use decisions to enable safe and effective care of all patients of all ages.

Regulatory guidelines for the ongoing monitoring of safety and quality metrics of AI model performance in clinical practice have not been established. However, changes in institutional demographics, such as the expansion of a neonatal intensive care unit or the acquisition of a new facility into a practice, hospital, or health system portfolio, can lead to the inclusion of different racial and ethnic demographics. The impact of such change on model performance is unknown and difficult to anticipate, requiring real-world reassessment and monitoring. Over time, such nuanced changes in populations served can result in model performance drift. Given the outcome of delays in diagnosis on the pediatric population, including on subsequent growth and development, especially considering the overall longer pediatric patient life expectancy, infrastructure and processes to monitor and maintain performance standards similar to those achieved during initial acceptance testing are more important in pediatric populations.

ECONOMICS

The lack of a well-defined structure for AI reimbursement remains a major challenge for the clinical deployment of AI, which is not unique to pediatrics. Practices are challenged to absorb the cost of AI licensing and deployment. Companies are investing significant venture capital in the development of product lines of ever more sophisticated AI solutions without clarity on how reimbursement for this technology will be woven into prevailing payment methodology. It remains to be seen if radiology AI will largely be a cost center or become associated with sustainable reimbursement for practices. Along those lines, the impact of AI on the professional work of radiologists will need to be clarified. Infrastructure investments by practices for AI implementation, including those needed to facilitate postdeployment AI performance monitoring, may need to become more

widespread to be properly captured in future practice expense surveys. This will be a prerequisite to properly consider this practice expense in any adjustments in the components of imaging reimbursement methodology.

To understand the specific challenge from a pediatric imaging perspective, only a small minority of pediatric patients (<2%) are covered by Medicare [27]. However, payment policy has been dominated by CMS for patients covered by Medicare [28]. Because Medicaid and the Children’s Health Insurance Program provide the majority of government support for pediatric patients, the reimbursement for pediatric care varies by state [29]. Finally, for nearly all pediatric patients who are government insured, the lower level of Medicaid reimbursements in similar settings for pediatric patients compared with Medicare for adults will also be relevant to access and use of AI for pediatric patients. For example, Medicaid fee-for-service payments are historically about 30% less than Medicare rates [30]. Providers reliant on revenue from government-insured pediatric patients would theoretically have less margin to invest in AI technology, an additional disincentive for companies to develop and market product lines for this population to health care providers.

To date, some AI product manufacturers for adult populations [28,31,32] have successfully achieved reimbursement through CMS for Medicare patients through an additional payment for new technology (the new technology add-on payment [NTAP]), which has been credited with increasing both the use of AI in medical imaging [33] and funding for the development of AI [34]. It is uncertain how the NTAP would apply to children.

Although the NTAP has set a precedence for payments, other insurers, including Medicaid and the Children’s Health Insurance Program and commercial payers, may or may not choose to follow this pathway [28]. The program applies only to devices that are substantially different from existing AI products. Therefore, even if other insurers adopt NTAP-like methodology for devices designed for pediatric patients after those for adults (ie, not considered innovative), this reimbursement mechanism may have limited or no applicability.

Current Procedure Terminology (CPT) coding is another method that has been used for AI reimbursement and may be more relevant than the NTAP in the long term. This reimbursement is influenced by the valuation process of the AMA/Specialty Society Relative Value Scale Update Committee (RUC), which makes recommendations to CMS regarding the relative value units (RVUs) for category I codes [28,35]. The RVUs of CPT codes are based on the time and intensity of the physician work involved in addition to the practice expense, which is composed of both direct and indirect costs. Direct costs include

nonphysician clinical labor and medical supplies, whereas indirect costs include administrative and overhead costs [26,33]. Because many AI products prioritize, enhance, or augment physician work, AI often represents an indirect cost. As such, this assistive AI may have a harder time qualifying for CPT codes through the RUC process. On the other hand, AI that is augmentative (ie, it produces output that normally cannot be performed by physicians) is more likely to qualify for a CPT code and associated payment.

Current work valuation is based on certain assumptions about physician effort related to typical complexity to interpret an examination while also taking historical complexities into consideration. Unless the documented physician effort, including time, substantially increases under conditions of simultaneous use of AI, it would probably not meet criteria for increasing the associated RVUs and, consequently, the reimbursement level of the corresponding CPT. As a result, in most current scenarios of patient care in which professional work is paid via existing CPT codes, AI would be a cost rather than a source of additional revenue. With added cost, the incentive for practices to use tools may decrease, especially if the designed tool can be used only among pediatric patients and not universally.

There are future complexities of AI on physician work that also need to be considered. Should AI eventually become predominantly or wholly used for normal studies, physician work intensity may increase, as physicians would be responsible for interpreting only the more complex or abnormal examinations.

The deployment of AI at scale in a clinical environment does require investment into infrastructure and support, and in addition to licensing fees, these costs represent an indirect practice expense [28]. Indirect costs do not typically qualify for CPT codes, as they have no impact on physician work time and intensity. Nonetheless, these increased costs of using AI in practices should be emphasized with the RUC to eventually achieve an appropriate upward adjustment in the technical component reimbursement. Should the reimbursement increase, there may be increased interest in developing tools that can also be used in or specifically for pediatric patients as well as the increased potential for widespread adoption and availability of pediatric AI. Payers could also then consider limited scope or budget payment programs for pediatric AI, perhaps in combination with a registry participation requirement analogous to the National Oncologic PET Registry, to drive future decision making about reimbursement [36]. This might incentivize use while limiting financial exposure to public payers and receiving real-world data on the frequency of use and efficacy in return.

RECOMMENDATIONS

Given the wide disparity between adult and pediatric AI development and availability in the market, near-term action is needed to ensure that AI becomes accessible, safe, effective, and reliable for pediatric patients. Recommendations are summarized in Table 1. In addition, the appropriate understanding and inclusion of pediatric radiologists in the field of AI is an essential step in developing mentorship for radiologists that care for children. Guidance from pediatric expert radiologists about the nuances of their practice is expected to play an important role in defining the development and adoption of pediatric AI technologies.

The promotion of pediatric AI development mitigates the potential negative outcomes for children due to uniformed and even wide and indiscriminate use of adult AI products. In most instances it is unknown whether AI designed for adults will work as intended and perform as reported in the specific indications for use. This is particularly relevant in practices in which both adult and pediatric patients are cared for. For example, if triage AI is used for worklist prioritization, pediatric patients will not benefit if

Table 1. Summarized recommendations to address current pediatric AI health equity issue

1. Include pediatric patients in AI models that are developed and potentially applicable to children.
2. Design effective and efficient methods for AI development, testing, and deployment among institutions, professional societies, and industry.
3. Advance a regulatory framework to promote consumer and public transparency.
4. Improve prioritization of pediatric AI research in funding agencies.
5. Engage patients and caregivers to understand priorities and needs for development.
6. Ensure equitable access to AI-enabled advances in health care.
7. Cultivate favorable health care economics incentivizing AI that improves patient outcomes.
8. Incorporate AI into clinical practice guidelines for children when appropriate.
9. Predetermine the level of radiologist supervision needed for output validation patient subgroups before clinical deployment.
10. Support collaboration with subspecialty content experts at all stages of AI design, development, implementation, and audit.

Note: AI = artificial intelligence.

Table 2. Summary of recent ACR government relations communications on pediatric AI

Communication	Recipient	Date	Key Requests
Transparency of AI/machine learning-enabled medical devices	FDA	November 15, 2021	■ Mandatory statement on intended age group (ie, pediatric, adult, or adult and pediatric) in all AI labels ■ Nutrition-style label with demographic information, including age
ACR/SPR response to request for information	US House of Representatives Healthy Future Task Force Treatments Subcommittee	March 11, 2022	■ Designate pediatric AI as breakthrough devices within FDA ■ Enhanced public access to information including mandatory pediatric use statement and nutrition label ■ Established research programs for pediatric AI ■ Report to Congress from secretary of HHS on a national strategy to advance safe and effective AI for pediatric patients
ACR comments on pediatric medical devices public-private partnership	National Institutes of Health (RFI NOT-EB-22-008)	September 21, 2022	■ Enable pediatric imaging data availability ■ Support validation or retraining of existing (adult) AI SaMD for use on pediatric patients ■ Fund new AI pediatric medical device research and development ■ Identify AI pediatric medical device payment models and financial incentives ■ Prioritize regulatory agency review and resources for AI pediatric medical devices ■ Promote AI transparency
ACR comments on section 1557 updates proposed rule; nondiscrimination in health programs and activities	HHS Office of Civil Rights (RIN 0945-AA17)	September 28, 2022	■ Coordinate with agencies that oversee AI to ensure that clinicians have access to all algorithm information necessary for compliance with the section 1557 “clinical algorithm use” proposal ■ Consider that AI tools may not always be safely and effectively generalizable across all section 1557-protected bases, such as patient age ■ Increased AI transparency, such as that recommended by the ACR to the FDA in 2021, is urgently needed to facilitate compliance with this proposal

Note: AI = artificial intelligence; HHS = US Department of Health and Human Services; SaMD = software as a medical device; SPR = Society for Pediatric Radiology.

similar urgency of their findings is not recognized by AI algorithms. Therefore, focused educational efforts across practice types on this potential impact are crucial. Additionally, safety measures such as appropriate filtering or separate pediatric worklists are likely needed to ensure that AI that has not been developed for use in pediatric patients does not negatively affect pediatric patient care.

The formation of the Pediatric AI Workgroup within the ACR Commission on Informatics represents a first step toward understanding the AI needs of pediatric populations [37]. Education and policy recommendations center on public access to algorithm information, improved funding and prioritization of AI development, and a national strategy for pediatric AI innovation, summarized in Table 2 [26,38-40]. The group maintains that improved public access to information about FDA-regulated AI products would enable health care providers to assess the safety and effectiveness profile and better care for their patients, especially children. Furthermore, the work group has established the Image IntelliGently™ campaign (www.imageintelligently.org) with the vision that pediatric patients' health will benefit from the use of AI in medical imaging. At the regional and national societal levels, the advocacy topic is gaining traction not only with the ACR and the Society for Pediatric Radiology but with partners outside radiology, such as the American Academy of Pediatrics [41].

CONCLUSIONS

Only a few FDA-authorized image-based AI algorithms have been developed specifically for children or adequately validated on pediatric cohorts. As a result, there is a critical need for an appropriate framework for pediatric AI development, validation, implementation, and audit. This currently unmet need could be addressed in several ways. First, the generalizability of existing adult AI algorithms should be assessed in pediatric cohorts before their potential use in children. Second, vendors could be incentivized to develop AI using suitable pediatric data, resulting either in separate pediatric models or in combined adult and pediatric data sets. Finally, regulatory bodies could encourage transparency about the inclusion of pediatric patients in AI data sets to ensure correct application of AI on the basis of customary metrics in pediatric care such as age. The potential failures of adult algorithms in pediatric settings, such as an AI algorithm missing a treatable pediatric disease, highlights the need to verify algorithms in the pediatric population.

The pediatric population cannot be approached as a homogeneous group of "little adults" and should not automatically be included in adult medical practices that rely upon AI devices not specifically designed or validated for

pediatric use without other workflow controls in place. AI tools should be specifically designed, trained, tested and validated, and monitored for safe and effective use in children. Similar to the Image Gently campaign in 2007, an alliance between concerned radiology organizations can serve to raise awareness of the need for improved implementation and governance of pediatric AI [4]. By providing information and free educational materials to every member of the care team, this group has demonstrated the ability to effectively advocate on behalf of pediatric patients with respect to informed use of medical radiation. On the eve of widespread use of AI in imaging practice, the radiology community should align to ensure fair and equitable access to clinically useful AI for pediatric patients and safeguard pediatric patient safety and Image IntelliGently.

TAKE-HOME POINTS

- To date, AI in radiology has been developed nearly exclusively for use in adult patients.
- Lack of access to AI is a health equity deficit for pediatric patients.
- AI developed for adults has not been shown to work consistently across the pediatric population.
- There are significant unique ethical concerns for the use and development of AI for pediatric patients.
- Current economic policies are largely not favorable for pediatric patients.
- Transparency in development processes is especially needed for pediatric patients.

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